

Chen-Yu Kuo

郭宸毓 • Systems & full-stack engineer — eBPF/Go infrastructure, ESP32 firmware, applied ML and LLM inference, earthquake early warning

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I build systems end to end, usually as the only author: an eBPF/XDP packet filter and a five-city Nginx/CoreDNS fleet written in Go, ESP32 firmware written register by register, Flutter and TypeScript clients, and PyTorch models I train, evaluate and quantise myself. Most of it runs under **ExpTech** (探索科技), the 29-person volunteer organisation I founded and one of two administrators of, which operates Taiwan’s largest independent earthquake early-warning platform. Alongside it I worked as a backend engineer at P-Waver, then as a part-time assistant at Academia Sinica’s Institute of Information Science, and now write software at the National Health Research Institutes.

Competences

Languages	TypeScript/JavaScript, Go, Python, Dart, C/C++, Rust, Lua, Shell
Infrastructure	Linux, eBPF/XDP, Docker, Nginx/OpenResty, CoreDNS, NTP/SNTP, Prometheus, CI/CD
Backend & apps	Go and Node/TypeScript services, REST/WebSocket/SSE, Flutter (iOS & Android), Electron, MapLibre
Embedded	ESP32/Arduino, FreeRTOS, I ² C/SPI, LSM6DS3, GNSS (QZSS L1S), LoRa/Meshtastic
Machine Learning	PyTorch, SeisBench, ONNX, scikit-learn, vLLM, llm-compressor, TensorRT Model Optimizer
Domain	Earthquake early warning, GMPE calibration, CWA/JMA intensity scales, phase picking
 Languages	Mandarin — native; English — technical reading & writing

Education

2022 – present	Dept. of Electrical Engineering, National Kaohsiung Univ. of Science & Technology B.S. in Electrical Engineering
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Experience

Jan 2025 – present	Software Developer — National Health Research Institutes (國家衛生研究院), Taiwan
Dec 2024 – Jun 2025	Part-time Assistant — Institute of Information Science, Academia Sinica (中央研究院資訊科學研究所), Taipei
Nov 2023 – Nov 2024	Backend Engineer — P-Waver Inc. (衛波科技, earthquake early-warning systems), Taipei
Oct 2020 – present	Founder & Lead Engineer — ExpTech (探索科技), Taiwan ➤ Lead a 29-member volunteer organisation across 234 repositories as one of its two administrators, spanning firmware, 37 Go services, Flutter clients and ML pipelines. Authored 13,411 commits inside it (19,284 across GitHub, 477 PRs) and am top contributor on every flagship codebase — DPIP 838/2,339, TREM-Lite 448/847, platform API 480/803 — and sole author of the 665-commit production infrastructure repository. ➤ Signed a data agreement with Taiwan’s Central Weather Administration (CWA) in late 2022 — reported by 親子天下 as the Seismological Center’s first non-corporate partner; invited back 10+ times as a collaborator (天下雜誌, Feb 2026). ➤ Ship under real stakes: 100,000+ Google Play installs and 4.2/5 from 1,378 App Store ratings for DPIP; 328,542 installer downloads across TREM desktop releases. During the M7.2 Hualien earthquake (3 Apr 2024) the network recorded 800+ events in a single day.

Security, Open Source & Press

- **Security & incident response** — operated a honeypot that captured in-the-wild exploitation of **CVE-2025-55182** (React Server Components, CVSS 10.0) within days of its December 2025 disclosure; documented six malware samples with verified hashes and fifteen attacker source addresses.
- **Open source** — 12 merged PRs to outside projects: a rendering-performance fix to the seismograph suite **AnyShake** (spectrogram-js), a Leaflet → MapLibre migration for smc.peering.tw, and Flutter upgrades to **NKUST-AP-Flutter**, my university’s student app.
- **Press** — 天下雜誌, Feb 2026 (quoted as an ExpTech collaborator in a CWA Seismological Center profile); 親子天下, Jul 2024 (DPIP feature, ~500,000 downloads after the 3 Apr 2024 quake).

Dec 2025 – present	Production infrastructure — packet filtering, edge and DNS Go, eBPF/XDP, OpenResty, CoreDNS, sole author ▶ kekkaï — packet filter in Go using cilium/ebpf, with an XDP ingress program and a TC egress hook; lru_hash/array/ringbuf maps drive per-IP rate limiting, allow/blocklists and port policy, with an emergency detach path. ▶ Nginx/OpenResty fleet — 23 upstreams across nodes in Taipei, Kaohsiung, Tainan, Tokyo and Osaka, with Lua modules for traffic accounting, NTP and image proxying. ▶ Split-horizon CoreDNS with a custom Go dnsstats plugin (miekg/dns) exposing per-domain/client query stats and Prometheus metrics. Sole author too of the NTP server, API gateway and Meshtastic LoRa ingest service.
Apr 2026 – Jul 2026	LLM inference & model compression Python, PyTorch, vLLM, TensorRT Model Optimizer, sole author ▶ Published 27 quantised models on Hugging Face (~73,000 all-time downloads) for vLLM and llama.cpp: NVFP4 via TensorRT Model Optimizer, INT4-AutoRound and GGUF, up to 35B mixture-of-experts. ▶ Implemented refusal-direction ablation — deriving the refusal direction from paired prompts and orthogonalising it out of the weight matrices, rather than steering at inference time — and released the <i>Ornith-1.0</i> 9B/35B line including a DPO-tuned variant.
Sep 2025 – Jun 2026	ES-Net — sensor firmware & 200-node device network C++, ESP32, FreeRTOS, sole author, 7,759 LOC ▶ Hand-wrote an LSM6DS3 accelerometer driver at register level instead of using a vendor library — own register map, who_am_i, ctrl1_xl, int1_ctrl, FIFO — plus an IIR filter bank and a binary indexed tree computing a running amplitude percentile on-device. ▶ Timing is the hard problem here: SNTP in smooth mode so the clock slews rather than steps mid-waveform, plus HW/SW watchdogs, OTA update, and a documented custom binary wire protocol with TOTP device authentication. ▶ Runs on 200+ nodes nationwide; first-generation units were self-assembled for under NT\$800 each.
Aug 2023 – present	DPiP — cross-platform mobile app & backend Dart/Flutter, TypeScript, top contributor, 838/2,339 commits ▶ iOS and Android client with push alerting, MapLibre layers and 10 interface languages; 100,000+ Google Play installs, 4.2/5 from 1,378 App Store ratings. Wrote the test suite and custom commit-gate tooling outright, and authored the TypeScript backend dpip-server-ts in full (122/122 commits).
Aug 2026 – present	QZSS L1S satellite message decoder C++, sole author, ~12k LOC with unit tests ▶ Decodes QZSS L1S 250-bit SBAS frames end to end — preamble, 6-bit message type, 212-bit payload and CRC-24Q — including type 43 DC Report (JMA disaster broadcasts, IS-QZSS-DCR) and type 44 DCX.
Mar 2026 – Jul 2026	Applied ML — regression & sequence models on sensor data Python, PyTorch, SeisBench, ONNX, sole author ▶ Refit a regression model on 45,398 CWA PGA/PGV/intensity observations (2000–2025), replacing an equation that over-amplified with magnitude and a site-response term that was double-counted, inflating estimates $\sim 2\times$. Held-out: 62.0% exact vs 39.7% for the model then in production, MAE 0.400 vs 0.671, 0% false alarms. ▶ Trained a 50 Hz PhaseNet sequence model over a 6,457-line pipeline (ingest, noise injection, evaluation, ONNX export). Diagnosed a windowing bug — the source window was shorter than the model window, so the model learned the event always sits mid-window and fired a false pick every 45 s stride.
Jan 2023 – present	TREM — desktop platform & third-party plugin ecosystem JavaScript, Electron, Rust, top contributor, 448/847 commits ▶ Cross-platform Electron client with 328,542 installer downloads across 74 releases, shipped continuously since Jan 2023. Designed the plugin system third parties extend it through and maintain its registry (417 commits); wrote the trem-monitor health service in Rust (105/150).